moment of separation. An average distribution can, however, be obtained by performing a large number of identical experiments. The number of decay events in each phase is recorded. The decay statistics are replaced by the statistical distribution of experimental results.

Studies using extremely small amounts of any shortlived radionuclide involve a number of statistical considerations. The binomial law has to be used since the conditions for the Poisson distribution, *i.e.* that $N > 100$, and that the number of observed decays, $\overline{N}_{\Delta t}$, in the measuring time, Δt, is much smaller than N, are seldom fulfilled in this case. The probability, $P(1)$, of exactly one event happening is according to the binomial distribution:

$$P(1) = \left(\frac{n!}{(n-1)!}\right) p(1-p)^{(n-1)} \tag{9.47}$$

where p is the probability of decay, and n is the total number of atoms.

For a radioactive decay, the probability for exactly one decay during the time Δt in a sample containing 1 atom is thus:

$$p(\Delta t) = 1 - \mathrm{e}^{-\lambda \Delta t} \tag{9.49}$$

However, when the half-life is long compared to Δt it is possible to simplify this expression by a truncated Taylor series expansion to:

$$p(\Delta t) = \lambda \Delta t \tag{9.50}$$

In the general case involving N atoms, the following expressions for the expected average of decayed atoms, $\overline{N}_{\Delta t}$, and its variance, σ, are obtained:

$$\overline{N}_{\Delta t} = N\left(1 - \mathrm{e}^{-\lambda \Delta t}\right) \tag{9.51}$$

and

$$\sigma = \sqrt{\overline{N}_{\Delta t}}(1 - \mathrm{e}^{-\lambda \Delta t}) \tag{9.52}$$

9.13. EXERCISES

9.1. A detector has a 1 cm^2 efficient area perpendicular to a γ-particle flux produced by a source 7 m away. The sensitivity for the 0.73 MeV γ-radiation is 8.2%. (a) What must the source strength be for the detector to register 1000 cpm? (b) What fraction of the radiation is absorbed in the air space between source and detector?

9.2. A ~100 MeV fission fragment is stopped in a plastic plate with density ~1 and an average atomic spacing of 0.25 nm. Estimate (a) the range in the plate, and (b) the

ionization density (ion pairs μm^{-1}). If the ionization along the track is spread out perpendicular from the track so that 1 in 10 atoms are ionized (c) what would be the diameter of the track? From the track dimensions (d) calculate the average energy deposition to each atom within the "cylinder", and, using the relation $E = 3\mathbf{k}T/2$, (e) estimate the average temperature within the track volume. In lack of basic data for the plastic material, use data for water.

9.3. Plutonium in a urine sample is soaked into a photographic emulsion so that the emulsion increases its volume by 20%. The 12 μm thick emulsion is dried to original thickness and then left in darkness for 24 h. After development, α-tracks are counted and an average of 2356 tracks cm^{-2} found. If the plutonium consists of 67% ^{239}Pu and 33% ^{240}Pu, what was the plutonium concentration in the urine?

9.4. A ^{244}Cm sample is measured in an ion chamber (Fig. 9.11). The voltage drop over a 3×10^{13} Ω resistor is measured to be 0.47 V. What is the activity of the sample if all α's emitted in the chamber (2π geometry) are stopped in the gas?

9.5. In a proportional counter filled with methane of 1 atm the gas multiplication is 2×10^4. What is the maximum pulse size for a 5.4 MeV α, if the ion-pair formation energy is assumed to be 30 eV? The capacitance of the circuit is 100 pF.

9.6. In a GM counter, sample A gave 12630, B 15480, and A + B together 25147 cpm. (a) What is the resolving time of the counter? (b) With the same counter, the distribution of radioactive samarium between an organic phase and water was measured according to $D_m = R_{org}/R_{aq}$. The measured R_{org} is 37160 cpm, and that of R_{aq} is 2965. (b) What is the measured D_m? (c) Using corrections for resolving time, what is the true D-value?

9.7. Assume that 10^9 alcohol molecules are dissociated per discharge in a GM tube of 100 cm^3 filled with 90% Ar and 10% ethyl alcohol vapour at a pressure of 100 mmHg (25°C). What is the lifetime of the tube in terms of total counts assuming this coincides with the dissociation of 95% of the alcohol molecules?

9.8. A 1 mm tick surface barrier detector of 10 mm diameter has a resistivity of 7000 Ω cm and a capacitance of 50 pF at 300 V reverse bias. Calculate the resolving time (time constant).

9.9. A plastic scintillation detector was to be calibrated for absolute measurements of β-radiation. For this purpose a 2.13×10^{-5} M $^{204}TlCl_3$ solution was available with a specific activity of 13.93 μCi ml^{-1}; ^{204}Tl emits β-particles with E_{max} 0.77 MeV. Of this solution 0.1 ml is evaporated over an area of exactly 0.1 cm^2 on a platinum foil. The sample is counted in an evacuated vessel at a distance of 15.3 cm from the detector, which has a sensitive area of 1.72 cm^2. The detector registers 2052 cpm with a background of 6 cpm. What is (a) the surface weight of the sample, (b) the backscattering factor, and (c) the detector efficiency for the particular β's?

9.10. A sample counted for 15 min gave 9000 total counts. A 30 min background measurement registered 1200 counts. Calculate (a) the count rate or the sample alone, with its standard deviation, and (b) with its probable error.

9.11. A certain sample has a true average counting rate of 100 cpm. What is the probability that 80 counts would be obtained in a 1 min recording?

9.14. LITERATURE

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